Exponential & Logarithmic Functions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • **Target: Multiplying and dividing rational expressions**

For the model $f(x) = 5x + 6$ in multiplying and dividing rational expressions, evaluate $f(5)$.

Final Answer: 31. Substitute $x = 5$: $f(5) = 5(5) + 6 = 31$.

2 [Foundational] • **Target: Multiplying and dividing rational expressions**

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

Final Answer: $x = 4$ and $x = 8$. Factoring $(x - 4)(x - 8) = 0$ yields the roots $x = 4$ and $x = 8$.

3 [Foundational] • **Target: Solving equations with rational expressions**

For the model $f(x) = 6x + 7$ in solving equations with rational expressions, evaluate $f(2)$.

Final Answer: 19. Substitute $x = 2$: $f(2) = 6(2) + 7 = 19$.

4 [Foundational] • **Target: Solving equations with rational expressions**

Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

Final Answer: $x = 1$ and $x = 6$. Factoring $(x - 1)(x - 6) = 0$ yields the roots $x = 1$ and $x = 6$.

5 [Foundational] • **Target: Solving equations with rational expressions**

A data set recorded for solving equations with rational expressions consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 19.5. The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

6 [Foundational] • **Target: Solving equations with rational expressions**

For the model $f(x) = 6x + 7$ in solving equations with rational expressions, evaluate $f(3)$.

Final Answer: 25. Substitute $x = 3$: $f(3) = 6(3) + 7 = 25$.

7 [Foundational] • **Target: Solving equations with rational expressions**

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.

Final Answer: $x = 2$ and $x = 7$. Factoring $(x - 2)(x - 7) = 0$ yields the roots $x = 2$ and $x = 7$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

© ClipSAT — clipsat.org | All Rights Reserved

8 [Intermediate] • Target: Solving equations with rational expressions

For the model $f(x) = 6x + 7$ in solving equations with rational expressions, evaluate $f(4)$.

Final Answer: 31. Substitute $x = 4$: $f(4) = 6(4) + 7 = 31$.

9 [Intermediate] • Target: Solving equations with rational expressions

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

Final Answer: $x = 3$ and $x = 8$. Factoring $(x - 3)(x - 8) = 0$ yields the roots $x = 3$ and $x = 8$.

10 [Intermediate] • Target: Solving equations with rational expressions

For the model $f(x) = 6x + 7$ in solving equations with rational expressions, evaluate $f(5)$.

Final Answer: 37. Substitute $x = 5$: $f(5) = 6(5) + 7 = 37$.

11 [Intermediate] • Target: Solving equations with rational expressions

Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

Final Answer: $x = 4$ and $x = 9$. Factoring $(x - 4)(x - 9) = 0$ yields the roots $x = 4$ and $x = 9$.

12 [Intermediate] • Target: Graphical and analytical solutions to rational equations

For the model $f(x) = 2x + 8$ in graphical and analytical solutions to rational equations, evaluate $f(2)$.

Final Answer: 12. Substitute $x = 2$: $f(2) = 2(2) + 8 = 12$.

13 [Intermediate] • Target: Graphical and analytical solutions to rational equations

A data set recorded for graphical and analytical solutions to rational equations consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 20.5. The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

14 [Intermediate] • Target: Graphical and analytical solutions to rational equations

For the model $f(x) = 2x + 8$ in graphical and analytical solutions to rational equations, evaluate $f(3)$.

Final Answer: 14. Substitute $x = 3$: $f(3) = 2(3) + 8 = 14$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Graphical and analytical solutions to rational equations • Cross-Lesson: 1.01 Functions & Their Graphs

For the model $f(x) = 2x + 8$ in graphical and analytical solutions to rational equations, evaluate $f(4)$.

Final Answer: 16. Substitute $x = 4$: $f(4) = 2(4) + 8 = 16$.

© ClipSAT — clipsat.org | All Rights Reserved

16 [Challenging] • Target: Graphical and analytical solutions to rational equations • Cross-Lesson: 10.01 Matrices & Systems

For the model $f(x) = 2x + 8$ in graphical and analytical solutions to rational equations, evaluate $f(5)$.

Final Answer: 18. Substitute $x = 5$: $f(5) = 2(5) + 8 = 18$.

17 [Challenging] • Target: Adding and subtracting rational expressions • Cross-Lesson: 11.01 Probability & Statistics

What is the sum of all real solutions to the equation $3^x + 3^{3-x} - 10 = |3^{x-1} - 3|$? Show your complete algebraic work.

Final Answer: $\log_3\left(\frac{81}{4}\right)$. To solve the equation, express all exponential terms in terms of a single substitution variable. Notice that: $3^x = 3 \cdot 3^{x-1}$ $3^{3-x} = \frac{27}{3^x} = \frac{27}{3 \cdot 3^{x-1}} = \frac{9}{3^{x-1}}$ Let $u = 3^{x-1}$. Since exponential functions with real inputs are strictly positive, we must have $u > 0$. The equation transforms into: $3u + \frac{9}{u} - 10 = |u - 3|$ Now consider the two cases determined by the absolute value: **Case 1: $u \geq 3$** Here.

18 [Challenging] • Target: Multiplying and dividing rational expressions • Cross-Lesson: 2.01 Quadratics & Complex Numbers

What is the sum of all real solutions to the equation $2^{\log_5(x^2)} - 6 \cdot x^{\log_5(2)} + 8 = 0$?.

Final Answer: 30. To solve the equation for real x, we first determine the domain and apply fundamental logarithmic identities. 1. **Domain:** The term $x^{\log_5(2)}$ requires $x > 0$ for the power to be well-defined over the real numbers. For $x > 0$, $\log_5(x^2)$ is also well-defined. 2. **Logarithmic and Exponential Identities:** Using the power rule for logarithms with $x > 0$: $\log_5(x^2) = 2\log_5(x)$ Thus, the first term can be rewritten as.

19 [Challenging] • Target: Solving equations with rational expressions • Cross-Lesson: 3.01 Polynomial Functions

Find all solutions of $\frac{x^2+2x}{x^2-1} = 1 + \frac{3}{x-1}$.

Final Answer: $x = 4$. LCD $= (x-1)(x+1)$: $x^2 + 2x = (x^2 - 1) + 3(x+1) \Rightarrow x^2 + 2x = x^2 + 3x + 2 \Rightarrow -x = 2 \Rightarrow x = -2$. Check: denom nonzero at $x = -2$. Answer: $x = -2$.

20 [Challenging] • Target: Graphical and analytical solutions to rational equations • Cross-Lesson: 4.01 Rational Expressions

Find all solutions of $\ln x + \ln(x-2) = 0$.

Final Answer: $x = 1 + \sqrt{2}$. $\ln(x(x-2)) = 0 \Rightarrow x(x-2) = 1 \Rightarrow x^2 - 2x - 1 = 0 \Rightarrow x = 1 + \sqrt{2}$ (positive, $x > 2$? $1 + 22.41 > 2$. ✓).

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

© ClipSAT — clipsat.org | All Rights Reserved

21 [Foundational] • Target: Simplifying rational expressions

For the model $f(x) = 3x + 4$ in simplifying rational expressions, evaluate $f(2)$.

a. 10 ✓

b. 13

c. 6

d. 14

Substitute $x = 2$: $f(2) = 3(2) + 4 = 10$.

22 [Foundational] • Target: Simplifying rational expressions

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

a. $x = 2$ and $x = 5$

b. $x = 2$ **and** $x = 4$ ✓

c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

Factoring $(x - 2)(x - 4) = 0$ yields the roots $x = 2$ and $x = 4$.

23 [Foundational] • Target: Simplifying rational expressions

For the model $f(x) = 3x + 4$ in simplifying rational expressions, evaluate $f(5)$.

a. 19 ✓

b. 22

c. 15

d. 23

Substitute $x = 5$: $f(5) = 3(5) + 4 = 19$.

24 [Foundational] • Target: Adding and subtracting rational expressions

For the model $f(x) = 4x + 5$ in adding and subtracting rational expressions, evaluate $f(2)$.

a. 13 ✓

b. 17

c. 8

d. 17

Substitute $x = 2$: $f(2) = 4(2) + 5 = 13$.

25 [Foundational] • Target: Adding and subtracting rational expressions

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.

a. $x = 2$ and $x = 6$

b. $x = 2$ **and** $x = 5$ ✓

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

Factoring $(x - 2)(x - 5) = 0$ yields the roots $x = 2$ and $x = 5$.

© ClipSAT — clipsat.org | All Rights Reserved

26 [Foundational] • Target: Adding and subtracting rational expressions

For the model $f(x) = 4x + 5$ in adding and subtracting rational expressions, evaluate $f(5)$.

a. 25 ✓

b. 29

c. 20

d. 29

Substitute $x = 5$: $f(5) = 4(5) + 5 = 25$.

27 [Foundational] • Target: Multiplying and dividing rational expressions

For the model $f(x) = 5x + 6$ in multiplying and dividing rational expressions, evaluate $f(2)$.

a. 16 ✓

b. 21

c. 10

d. 20

Substitute $x = 2$: $f(2) = 5(2) + 6 = 16$.

28 [Foundational] • Target: Multiplying and dividing rational expressions

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.

a. $x = 2$ and $x = 7$

b. $x = 2$ **and** $x = 6$ ✓

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

Factoring $(x - 2)(x - 6) = 0$ yields the roots $x = 2$ and $x = 6$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Simplifying rational expressions

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ **and** $x = 3$ ✓

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

Factoring $(x - 1)(x - 3) = 0$ yields the roots $x = 1$ and $x = 3$.

© ClipSAT — clipsat.org | All Rights Reserved

30 [Intermediate] • Target: Simplifying rational expressions

A data set recorded for simplifying rational expressions consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5 ✓

d. 18.5

The sum is $10 + 14 + 18 + 24 = 66$, and dividing by 4 gives 16.5.

31 [Intermediate] • Target: Simplifying rational expressions

For the model $f(x) = 3x + 4$ in simplifying rational expressions, evaluate $f(3)$.

a. 13 ✓

b. 16

c. 9

d. 17

Substitute $x = 3$: $f(3) = 3(3) + 4 = 13$.

32 [Intermediate] • Target: Simplifying rational expressions

For the model $f(x) = 3x + 4$ in simplifying rational expressions, evaluate $f(4)$.

a. 16 ✓

b. 19

c. 12

d. 20

Substitute $x = 4$: $f(4) = 3(4) + 4 = 16$.

33 [Intermediate] • Target: Simplifying rational expressions

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

a. $x = 3$ and $x = 6$

b. $x = 3$ **and** $x = 5$ ✓

c. $x = -3$ and $x = -5$

d. $x = 8$ and $x = 15$

Factoring $(x - 3)(x - 5) = 0$ yields the roots $x = 3$ and $x = 5$.

© ClipSAT — clipsat.org | All Rights Reserved

34 [Intermediate] • Target: Simplifying rational expressions

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ **and** $x = 6$ ✓

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

Factoring $(x - 4)(x - 6) = 0$ yields the roots $x = 4$ and $x = 6$.

35 [Intermediate] • Target: Adding and subtracting rational expressions

Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

a. $x = 1$ and $x = 5$

b. $x = 1$ **and** $x = 4$ ✓

c. $x = -1$ and $x = -4$

d. $x = 5$ and $x = 4$

Factoring $(x - 1)(x - 4) = 0$ yields the roots $x = 1$ and $x = 4$.

36 [Intermediate] • Target: Adding and subtracting rational expressions

A data set recorded for adding and subtracting rational expressions consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5 ✓

d. 19.5

The sum is $11 + 15 + 19 + 25 = 70$, and dividing by 4 gives 17.5.

37 [Intermediate] • Target: Adding and subtracting rational expressions

For the model $f(x) = 4x + 5$ in adding and subtracting rational expressions, evaluate $f(3)$.

a. 17 ✓

b. 21

c. 12

d. 21

Substitute $x = 3$: $f(3) = 4(3) + 5 = 17$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

© ClipSAT — clipsat.org | All Rights Reserved

38 [Challenging] • Target: Adding and subtracting rational expressions • Cross-Lesson: 2.01 Quadratics & Complex Numbers

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ **and** $x = 6$ ✓

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

Factoring $(x - 3)(x - 6) = 0$ yields the roots $x = 3$ and $x = 6$.

39 [Challenging] • Target: Adding and subtracting rational expressions • Cross-Lesson: 3.01 Polynomial Functions

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

a. $x = 4$ and $x = 8$

b. $x = 4$ **and** $x = 7$ ✓

c. $x = -4$ and $x = -7$

d. $x = 11$ and $x = 28$

Factoring $(x - 4)(x - 7) = 0$ yields the roots $x = 4$ and $x = 7$.

40 [Challenging] • Target: Multiplying and dividing rational expressions • Cross-Lesson: 4.01 Rational Expressions

Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

a. $x = 1$ and $x = 6$

b. $x = 1$ **and** $x = 5$ ✓

c. $x = -1$ and $x = -5$

d. $x = 6$ and $x = 5$

Factoring $(x - 1)(x - 5) = 0$ yields the roots $x = 1$ and $x = 5$.

41 [Challenging] • Target: Multiplying and dividing rational expressions • Cross-Lesson: 6.01 Radicals & Rational Exponents

A data set recorded for multiplying and dividing rational expressions consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5 ✓

d. 20.5

The sum is $12 + 16 + 20 + 26 = 74$, and dividing by 4 gives 18.5.

© ClipSAT — clipsat.org | All Rights Reserved

42 [Challenging] • Target: Multiplying and dividing rational expressions • Cross-Lesson: 7.01 Sequences & Series

For the model $f(x) = 5x + 6$ in multiplying and dividing rational expressions, evaluate $f(3)$.

a. 21 ✓

b. 26

c. 15

d. 25

Substitute $x = 3$: $f(3) = 5(3) + 6 = 21$.

43 [Challenging] • Target: Multiplying and dividing rational expressions • Cross-Lesson: 8.01 Conic Sections

For the model $f(x) = 5x + 6$ in multiplying and dividing rational expressions, evaluate $f(4)$.

a. 26 ✓

b. 31

c. 20

d. 30

Substitute $x = 4$: $f(4) = 5(4) + 6 = 26$.

44 [Challenging] • Target: Multiplying and dividing rational expressions • Cross-Lesson: 9.01 Trigonometric Functions

Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

a. $x = 3$ and $x = 8$

b. $x = 3$ **and** $x = 7$ ✓

c. $x = -3$ and $x = -7$

d. $x = 10$ and $x = 21$

Factoring $(x - 3)(x - 7) = 0$ yields the roots $x = 3$ and $x = 7$.

45 [Challenging] • Target: Solving equations with rational expressions • Cross-Lesson: 1.01 Functions & Their Graphs

A data set recorded for solving equations with rational expressions consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5 ✓

d. 21.5

The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

© ClipSAT — clipsat.org | All Rights Reserved